

Chapter 5

Uneven CashFlow

Question 1:

You are evaluating an investment opportunity offered by **Stellar Packaging Ltd.**, which involves a series of uneven annual cash inflows over the next five years. The company expects to generate the following cash flows from a new packaging machine investment:

Year	Cash Flow (PKR)
1	30,000
2	45,000
3	35,000
4	50,000
5	40,000

Assume the discount rate and reinvestment rate is 10% per annum.

- Calculate the Present Value (PV) of the uneven cash flows using a 10% discount rate.
- Calculate the Future Value (FV) of the same cash flows at the end of year 5, assuming all intermediate cash flows are reinvested at 10%.

Question 2:

You are considering three different investment opportunities—Project Alpha, Project Beta, and Project Gamma—each offered by company X.. These projects have uneven cash flows over a 4-year period. The company's required rate of return is 10% per annum.

The projected cash flows are as follows:

Year	Project Alpha (PKR)	Project Beta (PKR)	Project Gamma (PKR)
1	40,000	35,000	50,000
2	30,000	45,000	30,000
3	50,000	25,000	40,000
4	20,000	55,000	25,000

- Calculate the Present Value (PV) of cash flows for each project using a 10% discount rate.
- Calculate the Future Value (FV) at the end of year 4 for each project assuming all intermediate cash flows are reinvested at 10%.
- Based on your analysis, which project is financially most viable?

Question 3

An investment will pay \$150 at the end of next 3 years, \$250 at the end of year 4, \$300 at the end of year 5, and \$500 at the end of year 6. If other investments of equal risk earn 8% annually, what is the present value? And future value?

Question 4

Find the present values of the following cash flow streams at a 5% discount rate.

	0	1	2	3	4	5
Stream A	\$0	\$150	\$450	\$450	\$450	\$250
Stream B	\$0	\$250	\$450	\$450	\$450	\$150

Question no 5

A rookie quarterback is negotiating his first NFL contract. His opportunity cost is 7%. He has been offered three possible 4-year contracts. Payments are guaranteed, and they would be made at the end of each year. Terms of each contract are as follows:

	1	2	3	4
Contract 1	\$3,000,000	\$3,000,000	\$3,000,000	\$3,000,000
Contract 2	\$2,000,000	\$3,000,000	\$4,500,000	\$5,500,000
Contract 3	\$7,000,000	\$1,000,000	\$1,000,000	\$1,000,000

As his adviser, which contract would you recommend that he accept?